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SMA 2101: Calculus I

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This presentation is intended to covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green Exercise tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 2

Continuations of continuous functions

Any product or sum of continuous functions is continuous.

That is, if the functions f and g are continuous at $x = 2$, then

$$\lim_{x\to a}[f(x)+g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x) = f(a)+g(a)$$

Example $f(x) = x$

$f(x)$ is continuous everywhere

It follows that the cubic polynomial function $x^3 - 3x^2 + 1$ is continuous everywhere. More generally, every polynomial function $p(x) = b_n x^n + b_{n-1}x^{n-1} + \cdots + b_1 x + b_0$ is continuous at each point of the real line. If $p(x)$ and $g(x)$ are polynomials, then the quotient law for limits and the continuity of polynomials imply that $\lim\limits_{x\to a} \frac{p(x)}{g(x)} = \dfrac{\lim\limits_{x\to a} p(x)}{\lim\limits_{x\to a} g(x)} = \frac{p(a)}{g(a)}$, provided $g(a) \neq 0$.

Thus every rational function $f(x) = \frac{p(x)}{g(x)}$ is continuous wherever it is defined.

The point $x = a$ where the function f is discontinuous is called a removable discontinuity of f provided that there exist a function F such that

$F(x) = f(x)$ for all $x \neq a$ in the domain of f, and this new function F is continuous at $x = a$.

Example ✎. Suppose that $f(x) = \frac{x-2}{x^2-3x+2}$

Solution: but $x^2 - 3x + 2 = (x-2)(x-1)$

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$\therefore f(x) = \frac{x-2}{x^2-3x+2} = \frac{x-2}{x^2-3x+2} = \frac{1}{x-1}$. This shows that f is not defined at $x = 1$ and $x = 2$,

$\Longrightarrow f(x)$ is continuous except at these points. $f(x)$ is continuous except at the single point $x = 2$, the new function $F(x) = \frac{1}{x-1}$ agrees with $f(x)$ if $x \neq 2$ but it is continuous at $x = 2$ also where $F(2) = 1$. Therefore f has a removable discontinuity at $x = 2$. The discontinuity at $x = 1$ is not removable.

$\square$

Composition of functions

Let $h(x) = f(g(x))$ where f and g are continuous functions. The composition of two continuous functions is also continuous. More precisely if g is continuous at a and f is continuous at

$g(a)$, then $f \circ g$ is continuous at a where $f \circ g = f(g(x))$

Proof

The continuity of g at a means that $\lim\limits_{x \to a} g(x)$ and the continuity of f at $g(a)$ implies that $\lim\limits_{g(x) \to g(a)} f(g(x)) = f(g(a))$

$$\text{i.e.} \quad \lim_{x \to a} f(g(x)) = f\left(\lim_{x \to a} g(x)\right) = f(g(a))$$

Example. Show that the function $f(x) = \left(\frac{x-7}{x^2+2x+2}\right)^{2/3}$ is continuous on the whole real line.

Solution: Consider the denominator $x^2+2x+2 = (x+1)^2+1 > 0$ for all values of x, hence the rational function $r(x) = \frac{x-7}{x^2+2x+2}$ is defined and continuous everywhere. Thus $f(x) = \left(\frac{x-7}{x^2+2x+2}\right)^{2/3}$

is continuous everywhere □

More Revision Questions

(1). For the following problems find the points where the given function is not defined and therefore not continuous. For each given point a, tell whether this discontinuity is removable.

a) $f(x) = \frac{x-7}{(x+3)^2}$

b) $f(x) = \frac{x-2}{x^2-4}$

c) $f(x) = \frac{1}{1-|x|}$

d) $f(x) = \frac{x-17}{|x-17|}$

e) $f(x) = \begin{cases} -x & if \;\; x < 0 \\ x^2 & if \;\; x > 0 \end{cases}$

f) $f(x) = \begin{cases} 1 + x^2 & if \;\; x < 0 \\ \frac{\sin x}{x} & if \;\; x > 0 \end{cases}$

(2) For the following problems find a value of the constant c

so that the function $f(x)$ is continuous for all x.

a) $f(x) = \begin{cases} x + c & if \ x < 0 \\ 4 - x^2 & if \ x \geq 0 \end{cases}$

c) $f(x) = \begin{cases} c^2 - x^2 & if \ x < 0 \\ 2(x - c)^2 & if \ x \geq 0 \end{cases}$

b) $f(x) = \begin{cases} 2x + c & if \ x \leq 3 \\ 2c - x & if \ x > 3 \end{cases}$

d) $f(x) = \begin{cases} c^2 - x^3 & if \ x \leq \pi \\ c \sin x & if \ x > \pi \end{cases}$

2.1. THE DERIVATIVE OF A FUNCTION

Definition: The derivative of a function f is the function f' defined by;

$f'(x) = \lim_{h \to 0} \left[\frac{f(x+h)-f(x)}{h}\right]$ for all x for which this limit exists.

The function f is differentiable at $x = a$ if $\lim_{x \to a} f(x) = f(a)$ exist.

The process of finding the derivative f' is called differentiation of f

Example ✎. Find the derivative of the following from first principles

$f(x) = x^2$

Solution: $f(x) = x^2$

$$f'(x) = \lim_{h\to 0}\left[\frac{f(x+h)-f(x)}{h}\right]$$

$$f(x) = x^2, \qquad f(x+h) = (x+h)^2 = x^2 + 2xh + h^2$$

$$\therefore f'(x) = \lim_{h\to 0}\left[\frac{(x^2+2xh+h^2)-x^2}{h}\right] = \lim_{h\to 0}\left[\frac{2xh-h^2}{h}\right] = \lim_{h\to 0} 2x-h = 2x$$

$\square$

Example ✎. $f(x) = \sqrt{x}$

Solution: $f(x) = \sqrt{x}$

$$f'(x) = \lim_{h\to 0}\left[\frac{f(x+h)-f(x)}{h}\right]$$

$$f(x) = \sqrt{x}, \qquad f(x+h) = \sqrt{x+h}$$

$$\therefore f'(x) = \lim_{h\to 0}\left[\frac{\sqrt{x+h}-\sqrt{x}}{h}\right] = \lim_{h\to 0}\left[\frac{\left(\sqrt{x+h}-\sqrt{x}\right)\left(\sqrt{x+h}+\sqrt{x}\right)}{h\left(\sqrt{x+h}+\sqrt{x}\right)}\right]$$

$$= \lim_{h\to 0}\frac{x+h-x}{h\left(\sqrt{x+h}+\sqrt{x}\right)} = \lim_{h\to 0}\frac{h}{h\left(\sqrt{x+h}+\sqrt{x}\right)} == \lim_{h\to 0}\frac{1}{\left(\sqrt{x+h}+\sqrt{x}\right)} = \frac{1}{2\sqrt{x}}$$

$\square$

Example ✎. $f(x) = {}^1/_x$

Solution: $f(x) = {}^1/_x$

$$f'(x) = \lim_{h\to 0}\left[\frac{f(x+h)-f(x)}{h}\right]$$

$$f(x) = {}^1/_x, \qquad f(x+h) = {}^1/_{x+h}$$

$$\therefore f'(x) = \lim_{h\to 0}\left[\frac{{}^1/_{x+h}-{}^1/_x}{h}\right] = \lim_{h\to 0}\left[\frac{x-(x+h)}{h(x(x+h))}\right]$$

$$= \lim_{h \to 0} \left[\frac{-h}{h(x^2+xh)}\right] = \lim_{h \to 0} \left[\frac{-1}{(x^2+xh)}\right] = {}^{-1}\!/_{x^2}$$

$\square$

EXERCISE 1. ✍ $f(x) = x^n, \quad n > 0$

More Revision Questions

Differentiate the following functions from the first principles

1. $f(x) = x^3$
2. $f(x) = x^2 + 3x - 2$
3. $f(x) = 4x^4$

NB. Differential notation;

$$\Delta x = h, \qquad \Delta y = f(x + \Delta x) - f(x), \qquad \frac{\Delta y}{\Delta x} = \frac{f(x+\Delta x)-f(x)}{h}$$

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

If $y = f(x)$, we often write $\frac{dy}{dy} = f'(x)$

Solutions to Exercises

Exercise 1. $f(x) = x^n, \quad n > 0$

$$f'(x) = \lim_{h\to 0}\left[\frac{f(x+h)-f(x)}{h}\right]$$

$f(x) = x^n, \qquad f(x+h) = (x+h)^n = x^n + nx^{n-1}h + \frac{n(n-1)}{2!}x^{n-2}h^2 + \cdots + h^n$

$$\therefore f'(x) = \lim_{h\to 0}\left[\frac{\left(x^n+nx^{n-1}h+\frac{n(n-1)}{2!}x^{n-2}h^2+\cdots+h^n\right)-x^n}{h}\right]$$

$$= \lim_{h\to 0}\left[\frac{nx^{n-1}h+\frac{n(n-1)}{2!}x^{n-2}h^2+\cdots+h^n}{h}\right]$$

$$= \lim_{h\to 0} nx^{n-1} + \frac{n(n-1)}{2!}x^{n-2}h + \cdots + h^{n-1} = nx^{n-1}$$

Exercise 1

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